

Study Objective

The objective of this study was to evaluate vegetation change before and after the Morgan Creek Fire in Routt County, Colorado using the Normalized Difference Vegetation Index (NDVI). The analysis examines whether areas near mapped beaver pond clusters show different vegetation recovery patterns compared to the surrounding burned landscape.

Data Sources

Several spatial datasets were used in this analysis. Landsat satellite imagery was obtained from the USGS EarthExplorer platform. A pre-fire image from June 10, 2021 was selected because it occurred shortly before the fire and had 0.64 percent cloud cover. A post fire image from June 21, 2022 was selected to represent vegetation conditions approximately one year after the fire and had 0.13 percent cloud cover.

The Morgan Creek Fire perimeter was obtained through the [National Interagency Fire Center](#)

I obtained the “Upper Yampa Basin” watershed polygon from the [USGS Watershed Boundary Dataset](#).

Beaver pond cluster data was obtained from the Colorado Beaver Activity Mapper (COBAM), which compiles observations of beaver activity derived from the Beaver Restoration Assessment Tool (BRAT) and field observations.

Analytical Methods and Workflow

After I repaired the geometry for the fire polygon, calculating NDVI went smoothly. NDVI was calculated in ArcGIS Pro using the Raster Calculator with Landsat bands 5 (near infrared) and 4 (red), for both pre and post fire layers

Higher NDVI values represent healthier vegetation while lower values indicate sparse vegetation, burned areas, or bare soil.

I created a 200ft buffer around the beaver ponds and ran zonal statistics as table to find the mean value of NDVI pre and post fire for the beaver pond buffers.

Results

The NDVI maps show noticeable differences in vegetation conditions before and after the Morgan Creek Fire. The pre fire NDVI map from June 2021 shows generally moderate to high

vegetation productivity throughout the study area, represented by yellow and green tones. These colors indicate areas of healthy vegetation and dense plant cover prior to the fire.

The post fire NDVI map from June 2022 shows a reduction in vegetation productivity across much of the burned area. Many areas within the fire perimeter appear in orange and red tones, indicating lower NDVI values associated with burned or sparsely vegetated landscapes.

However, a noticeable patch of green vegetation remains near several mapped beaver pond clusters. These areas occur primarily along riparian corridors and appear greener than the surrounding burned landscape.

Zonal statistics further support this pattern. Within the 200 foot buffer surrounding the beaver pond clusters, the average NDVI value increased by (+0.067) after the fire. In contrast, the average NDVI value across the entire Morgan Creek Fire perimeter decreased by (−0.064) after the fire.

Overall, these results suggest that while vegetation productivity declined across most of the burned landscape, areas near beaver pond clusters maintained or slightly increased vegetation greenness relative to the broader fire area.

Discussion

The results suggest that riparian areas associated with beaver pond clusters may support stronger vegetation recovery following wildfire disturbance. Beaver ponds slow and store water, which increases groundwater storage and helps maintain higher soil moisture in surrounding floodplain environments. These wetter conditions can create riparian zones that are more resilient to disturbance and may allow vegetation to recover more quickly after wildfire.

The presence of greener vegetation patches near mapped beaver pond clusters in the post fire NDVI imagery supports the idea that beaver influenced landscapes can contribute to ecological resilience following wildfire events. While much of the burned landscape shows lower NDVI values, these localized riparian areas appear to retain higher vegetation productivity relative to surrounding terrain. This pattern suggests that beaver modified landscapes may function as small refugia where vegetation persists or recovers faster after fire.

However, several limitations should be considered. This analysis relies on imagery from only two time periods, which limits the ability to evaluate longer term vegetation recovery patterns. Including additional years of satellite imagery would allow for a more complete understanding of how vegetation changes through time after the fire.

In addition, NDVI measures vegetation greenness but does not identify vegetation species or structural changes in plant communities. Areas that appear greener may represent early regrowth or different vegetation types compared to pre fire conditions.

Despite these limitations, this analysis demonstrates how remote sensing tools such as NDVI can help identify spatial patterns of post fire recovery and highlight the potential ecological influence of beaver activity within fire affected landscapes.